

Supporting Information: Metric-consistent neural networks for genomic prediction and selection

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1 Scope and frozen outputs

The confirmatory benchmark comprises 101 public trait–dataset tasks from 12 panels, seven neural architectures, four losses, ten cross-validation test folds and three calibration maps, plus GBLUP and ridge baselines. The assembled file `results/results_main_campaign.parquet` contains 90,900 unique rows. The nested sensitivity file `results/results_loss_specific_wheat.parquet` contains 1,080 rows. No private dataset is used. The analysis validates row counts, experimental keys and task coverage before writing `results/analysis_final/`.

The main campaign has 588 missing Pearson/Spearman values, 270 missing R^2 and NDCG@10 values, and 334 missing calibration slopes/intercepts. Missing correlations are concentrated in nearly constant folds of the rice seed-color trait. RMSE is complete. Aggregation uses available finite folds and reports the resulting task count.

2 Architecture adaptations

The named networks test published motifs under a common protocol and are not exact reproductions.

Table 1: Relationship between the published systems and benchmark adaptations.

Model	Retained motif	Protocol adaptation
DeepGS (Ma et al., 2018)	marker convolution and dense prediction head	common dosage input and training loop; batch normalization added
DNNGP (Wang et al., 2023)	PCA-informed convolutional processing	genotype-only rather than the complete multi-omic setting; common validation protocol
PNNGS (Xie et al., 2024)	parallel residual kernels	strided stem and adaptive pooling; no reproduction of the published stratification protocol
SoyDNNGP (Gao et al., 2023)	two-dimensional convolution of reshaped markers	96×96 one-channel dosage image with repetition/truncation rather than 206×206×3 encoding

3 Estimands and inference

Raw Pearson is primary, raw NDCG@10 secondary, and affine NRMSE is the principal scale outcome. NRMSE divides RMSE by the phenotype SD in the corresponding test fold. Folds are averaged first. For a pooled effect, architectures are averaged within task, then the 12 panel means receive equal weight. Intervals are based on 20,000 panel-cluster bootstrap samples. Exact sign-flip tests reverse every task within a panel together. Trait-weighted and leave-one-panel-out estimates are available in the machine-readable table.

Table 2: Panel-balanced raw-Pearson contrasts relative to MSE. Intervals resample panels; the exact test flips signs at panel level.

loss	model	n_tasks	n_panels	mean_delta	ci_lo	ci_hi	panel_signflip_p	win_fraction
pearson	pooled	101	12	0.0048	0.0008	0.0091	0.0522	0.7723
pearson	mlp	101	12	0.0044	0.0011	0.0083	0.0273	0.6238

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loss	model	n_tasks	n_panels	mean_delta	ci_lo	ci_hi	panel_signflip_p	win_fraction
pearson	cnn	101	12	0.0008	-0.0102	0.0131	0.8994	0.4950
pearson	transformer	101	12	0.0176	0.0041	0.0370	0.0195	0.7327
pearson	deepgs	101	12	-0.0067	-0.0139	-0.0008	0.0752	0.3762
pearson	dnngp	101	12	0.0022	-0.0027	0.0074	0.4238	0.6040
pearson	pnngs	101	12	0.0096	-0.0031	0.0237	0.2095	0.5941
pearson	soydnngp	101	12	0.0056	-0.0018	0.0140	0.2139	0.5644
hybrid	pooled	101	12	0.0055	0.0010	0.0100	0.0425	0.8020
hybrid	mlp	101	12	0.0023	-0.0001	0.0050	0.1309	0.6040
hybrid	cnn	101	12	-0.0002	-0.0102	0.0118	0.9751	0.4950
hybrid	transformer	101	12	0.0191	0.0043	0.0388	0.0308	0.7228
hybrid	deepgs	101	12	-0.0030	-0.0071	0.0005	0.2061	0.4653
hybrid	dnngp	101	12	0.0011	-0.0027	0.0050	0.6050	0.6139
hybrid	pnngs	101	12	0.0091	-0.0049	0.0234	0.2607	0.6040
hybrid	soydnngp	101	12	0.0101	0.0033	0.0175	0.0273	0.6139
ccc	pooled	101	12	0.0078	0.0022	0.0139	0.0278	0.7525
ccc	mlp	101	12	0.0029	-0.0012	0.0074	0.2344	0.6337
ccc	cnn	101	12	0.0069	-0.0031	0.0186	0.2881	0.5644
ccc	transformer	101	12	0.0201	0.0054	0.0395	0.0215	0.7129
ccc	deepgs	101	12	-0.0037	-0.0071	-0.0007	0.0610	0.4158
ccc	dnngp	101	12	0.0068	0.0001	0.0138	0.0894	0.6733
ccc	pnngs	101	12	0.0014	-0.0105	0.0110	0.8145	0.5248
ccc	soydnngp	101	12	0.0204	0.0102	0.0324	0.0005	0.7624

Table 3: Panel-balanced pooled contrasts for all prespecified end-points.

metric	calibration	loss	n_tasks	mean_delta	ci_lo	ci_hi	panel_signflip_p
pearson	raw	pearson	101	0.0048	0.0008	0.0091	0.0522
pearson	raw	hybrid	101	0.0055	0.0010	0.0100	0.0425
pearson	raw	ccc	101	0.0078	0.0022	0.0139	0.0278
spearman	raw	pearson	101	0.0037	-0.0001	0.0078	0.1094
spearman	raw	hybrid	101	0.0047	0.0005	0.0090	0.0625
spearman	raw	ccc	101	0.0068	0.0017	0.0123	0.0288
ndcg@10	raw	pearson	101	0.0018	-0.0001	0.0035	0.0835
ndcg@10	raw	hybrid	101	0.0022	0.0003	0.0038	0.0444
ndcg@10	raw	ccc	101	0.0034	0.0015	0.0056	0.0078
overlap@10	raw	pearson	101	0.0033	-0.0010	0.0069	0.1494
overlap@10	raw	hybrid	101	0.0020	-0.0010	0.0046	0.2075
overlap@10	raw	ccc	101	0.0050	0.0019	0.0085	0.0112
releff@10	raw	pearson	101	0.0053	0.0007	0.0104	0.0605
releff@10	raw	hybrid	101	0.0052	0.0003	0.0093	0.0562
releff@10	raw	ccc	101	0.0086	0.0039	0.0137	0.0063
rmse	affine	pearson	101	0.0324	-0.0058	0.0928	0.2705
rmse	affine	hybrid	101	0.0227	-0.0055	0.0585	0.2773
rmse	affine	ccc	101	-0.0018	-0.0281	0.0335	0.9360
nrmse	affine	pearson	101	-0.0011	-0.0036	0.0015	0.4097
nrmse	affine	hybrid	101	-0.0018	-0.0041	0.0006	0.1616

nrmse	affine	ccc	101	-0.0028	-0.0054	-0.0003	0.0630
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4 Corroborative interaction models

The mixed model uses panel as a fixed blocking factor and a random intercept for trait–dataset task. Full models contain loss-by-architecture interaction terms; likelihood-ratio tests compare them with additive models. The interaction is not confirmed at 5% for any of the three endpoints, so architecture-specific intervals are interpreted descriptively.

Table 4: Likelihood-ratio tests for the loss-by-architecture interaction in corroborative mixed models.

metric	tasks	panels	LR	df	p	task variance
pearson	101	12	12.4043	12	0.4138	0.0001
ndcg@10	101	12	20.1917	12	0.0635	0.0000
nrmse	101	12	19.6731	12	0.0735	0.0000

5 Calibration

The theoretical rank-preservation result requires a positive affine slope. The benchmark used unconstrained least squares fitted on validation predictions. Of 7,014 Pearson-trained fold cells with finite raw and affine correlation, 12 (0.17%) changed sign. Raw predictions are used for the primary correlation and upper-tail loss comparisons; the headline figure additionally audits raw versus affine correlation to assess retention of the gain.

Table 5: Calibration audit for Pearson-trained networks. Raw-unit means are descriptive across heterogeneous traits.

map	n	median r	median RMSE	median slope	sign flips	flip fraction
raw	7070	0.5557	1.8159	0.9176	12	0.0017
affine	7070	0.5578	1.2850	0.9567	12	0.0017
isotonic	7070	0.5351	1.3215	0.8228	12	0.0017

6 Loss-specific tuning and initialization

For each of MLP, CNN and Transformer and each of five outer folds, MSE and Pearson used identical banks of eight candidates and two inner folds. Selection used Pearson on environment-1 outer-training individuals; the winner was then evaluated on four wheat environments with seeds 0, 1 and 2. Thus the four outcomes are transfers from one tuning phenotype, not four independently tuned experiments. Only selected configurations and the nested trial ledger are retained; the main campaign retains selected configurations but not its complete trial ledger.

Table 6: Loss-specific nested-HPO sensitivity on CIMMYT wheat; repeated outcomes are summarized descriptively.

metric	model	mean delta	minimum	maximum	pairs
pearson	pooled	0.0093	-0.0121	0.0785	180
pearson	mlp	0.0038	-0.0025	0.0097	60
pearson	cnn	-0.0044	-0.0119	0.0001	60
pearson	transformer	0.0286	-0.0121	0.0785	60
ndcg@10	pooled	0.0027	-0.0345	0.0377	180
ndcg@10	mlp	0.0071	0.0016	0.0151	60

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Table 6: Loss-specific nested-HPO sensitivity on CIMMYT wheat; repeated outcomes are summarized descriptively.

metric	model	mean delta	minimum	maximum	pairs
ndcg@10	cnn	-0.0060	-0.0198	0.0128	60
ndcg@10	transformer	0.0069	-0.0345	0.0377	60
nrmse	pooled	-0.0012	-0.0312	0.0124	180
nrmse	mlp	-0.0015	-0.0038	0.0036	60
nrmse	cnn	0.0073	0.0037	0.0124	60
nrmse	transformer	-0.0096	-0.0312	0.0011	60

7 Absolute method ranks

Ranks compare all 30 methods within each task using raw predictions and then average the 101 tasks with equal trait weight (unlike the panel-balanced primary estimand). Ridge and GBLUP rank first and second on average; neural-loss comparisons should therefore be understood as within-architecture improvements rather than superiority over linear genomic prediction.

Table 7: Mean ranks over 101 tasks using raw predictions.

method	ndcg@10	pearson
ridge	7.94	4.71
gblup	8.95	6.03
deepgs/mse	9.52	9.56
mlp/pearson	11.30	9.69
mlp/hybrid	11.29	9.82
mlp/ccc	11.21	10.07
deepgs/hybrid	10.11	10.30
mlp/mse	11.51	11.04
deepgs/ccc	11.13	11.28
soydnngp/ccc	10.69	11.30
deepgs/pearson	10.93	11.55
soydnngp/hybrid	13.36	13.07
soydnngp/pearson	14.07	13.23
cnn/ccc	15.71	14.45
cnn/pearson	14.93	15.23
soydnngp/mse	15.08	15.37
cnn/hybrid	15.28	15.77
dnngp/ccc	13.80	15.99
cnn/mse	16.34	16.38
dnngp/pearson	14.34	16.44
dnngp/hybrid	14.86	16.94
dnngp/mse	15.51	17.84
transformer/ccc	21.25	20.68
transformer/hybrid	20.77	21.07
transformer/pearson	20.98	21.10
pnngs/ccc	25.37	24.71
pnngs/mse	25.01	24.86
transformer/mse	23.56	25.23
pnngs/hybrid	24.95	25.29
pnngs/pearson	25.25	26.01

8 Figure source data and reproducibility

Analysis and simulation scripts are provided in Supplementary File 2; figure source tables and text alternatives are provided in Supplementary File 3. Every main-text figure is exported as PDF, SVG and PNG. Machine-readable panel data are in `figures/source_data/`. Reproduction commands are:

- Primary analysis: `python analysis/final_analysis.py`
- Interaction models: `python analysis/mixed_model_final.py`
- Tables: `python analysis/make_final_supplement.py`
- Figures: `python figures/make_figures.py`

The main benchmark uses one neural initialization per fold; only the wheat sensitivity quantifies three-seed variation.

References

- Gao, P., Zhao, H., Luo, Z., Lin, Y., Feng, W., Li, Y., et al. (2023). SoyDNGP: a web-accessible deep learning framework for genomic prediction in soybean breeding. *Briefings in Bioinformatics*, 24(6):bbad349.
- Ma, W., Qiu, Z., Song, J., Li, J., Cheng, Q., Zhai, J., and Ma, C. (2018). A deep convolutional neural network approach for predicting phenotypes from genotypes. *Planta*, 248(5):1307–1318.
- Wang, K., Abid, M. A., Rasheed, A., Crossa, J., Hearne, S., and Li, H. (2023). DNNGP, a deep neural network-based method for genomic prediction using multi-omics data in plants. *Molecular Plant*, 16(1):279–293.
- Xie, Z., Weng, L., He, J., Feng, X., Xu, X., Ma, Y., Bai, P., and Kong, Q. (2024). PNNGS, a multi-convolutional parallel neural network for genomic selection. *Frontiers in Plant Science*, 15:1410596.